PROGRAMMING FUNDAMENTALS

LAB 02 SUBMISSION

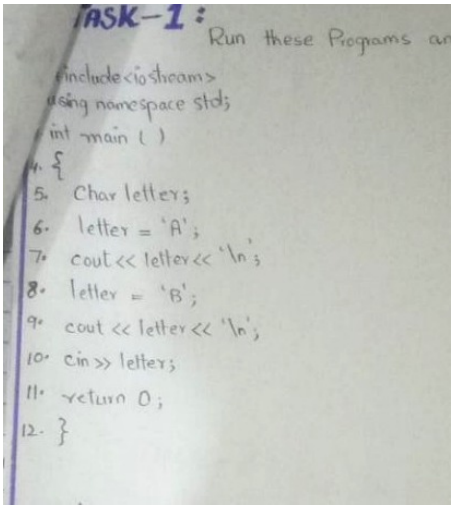
C++ Programs, Handwritten Work, Code and Output

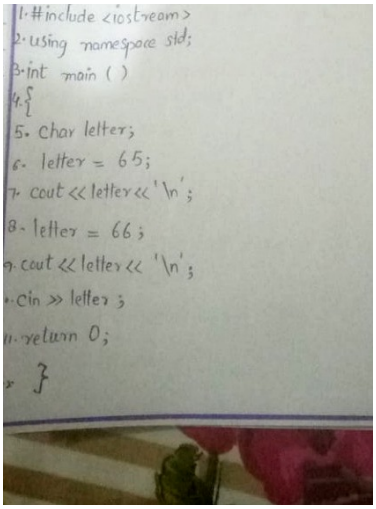
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Program	BS Computer Science (BSCS)
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Submitted To	Sir Asad
Date	12 October 2026

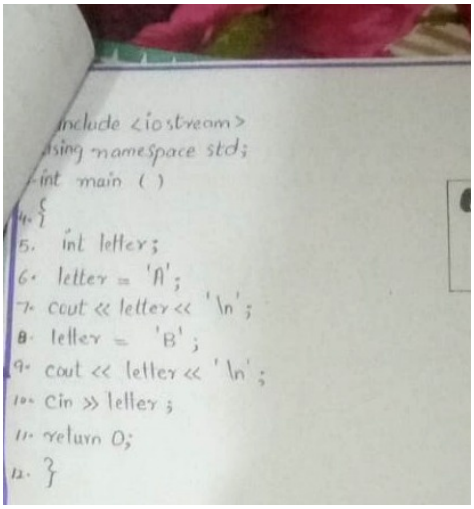
Learning variables, input/output and basic arithmetic in C++

Task 1 — Character Variables

Run the given programs and observe their output. Handwritten code crops are kept in the same order as your notebook.







Task 1 — Code & Output

Code in Compiler

```
#include <iostream>
using namespace std;
int main() {
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    return 0;
```

Output

```
A
B
```

```
#include <iostream>
using namespace std;
int main() {
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    return 0;
```

Output

```
A
B
```

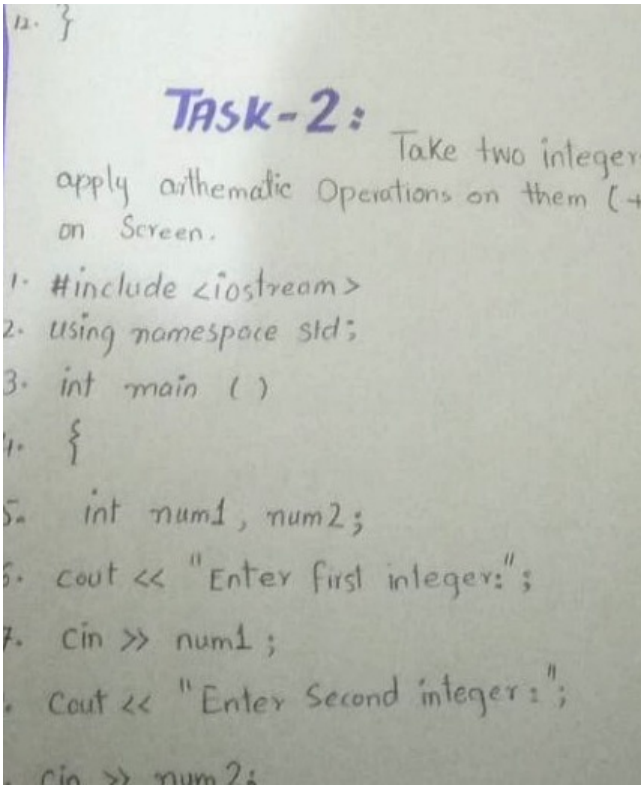
```
#include <iostream>
using namespace std;
int main() {
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    return 0;
```

Output

```
65
66
```

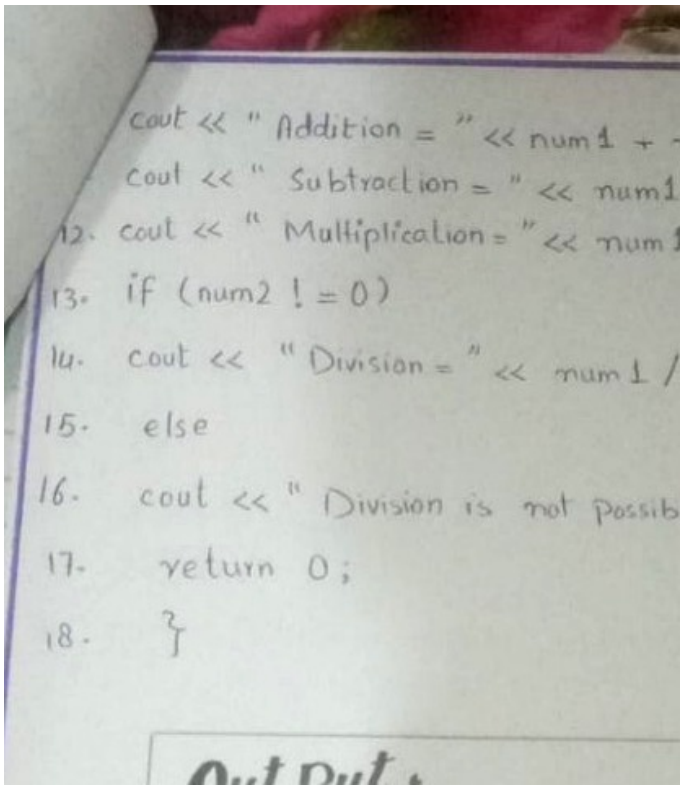
Task 2 — Arithmetic Operations

Take two integers as input from the user and display addition, subtraction, multiplication and division.



```
12. }

TASK-2: Take two integer
apply arithematic Operations on them (+
on Screen.
1. #include <iostream>
2. using namespace std;
3. int main ()
4. {
5.  int num1, num2;
6. cout << "Enter first integer:";
7. cin >> num1;
. Cout << "Enter Second integer:";
. cin >> num2;
```



```
cout << " Addition = " << num1 +
cout << " Subtraction = " << num1
12. cout << " Multiplication = " << num
13. if (num2 != 0)
14. cout << " Division = " << num1 /
15.  else
16.  cout << " Division is not possib
17.  return 0;
18. }

Out Put :
```

Task 2 — Code & Output

Code in Compiler

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b;
    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Addition = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;
    if (b != 0)
        cout << "Division = " << (double)a / b << endl;
    else
        cout << "Division is not possible (divisor is zero)." << endl;

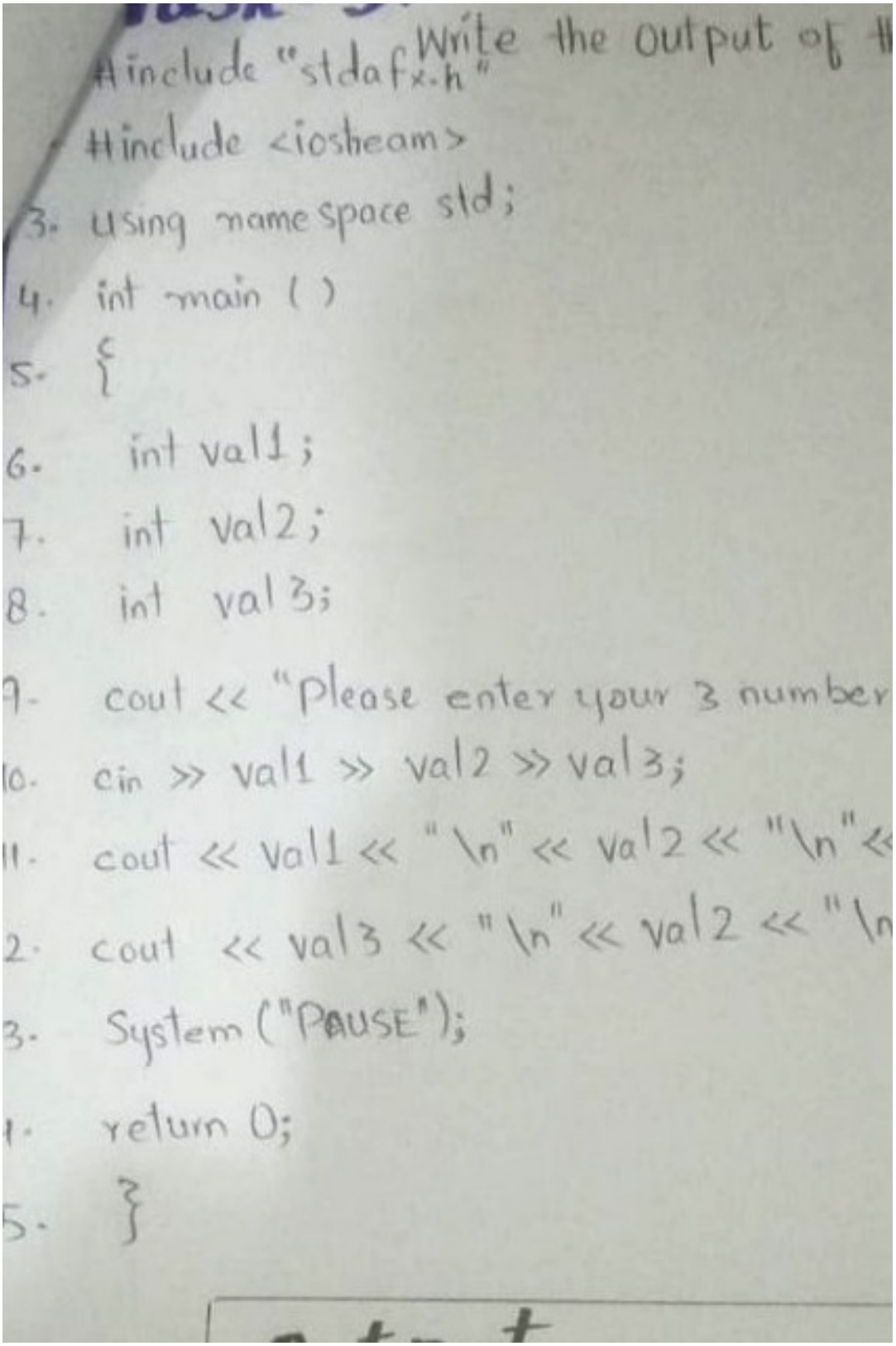
```

Output

```
Enter two integers: 12 4
Addition = 16
Subtraction = 8
Multiplication = 48
Division = 3
```

Task 3 — Write the Output

Enter three numbers and display them in the original order and then in reverse order.



Task 3 — Code & Output

Code in Compiler

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1, val2, val3;
    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;

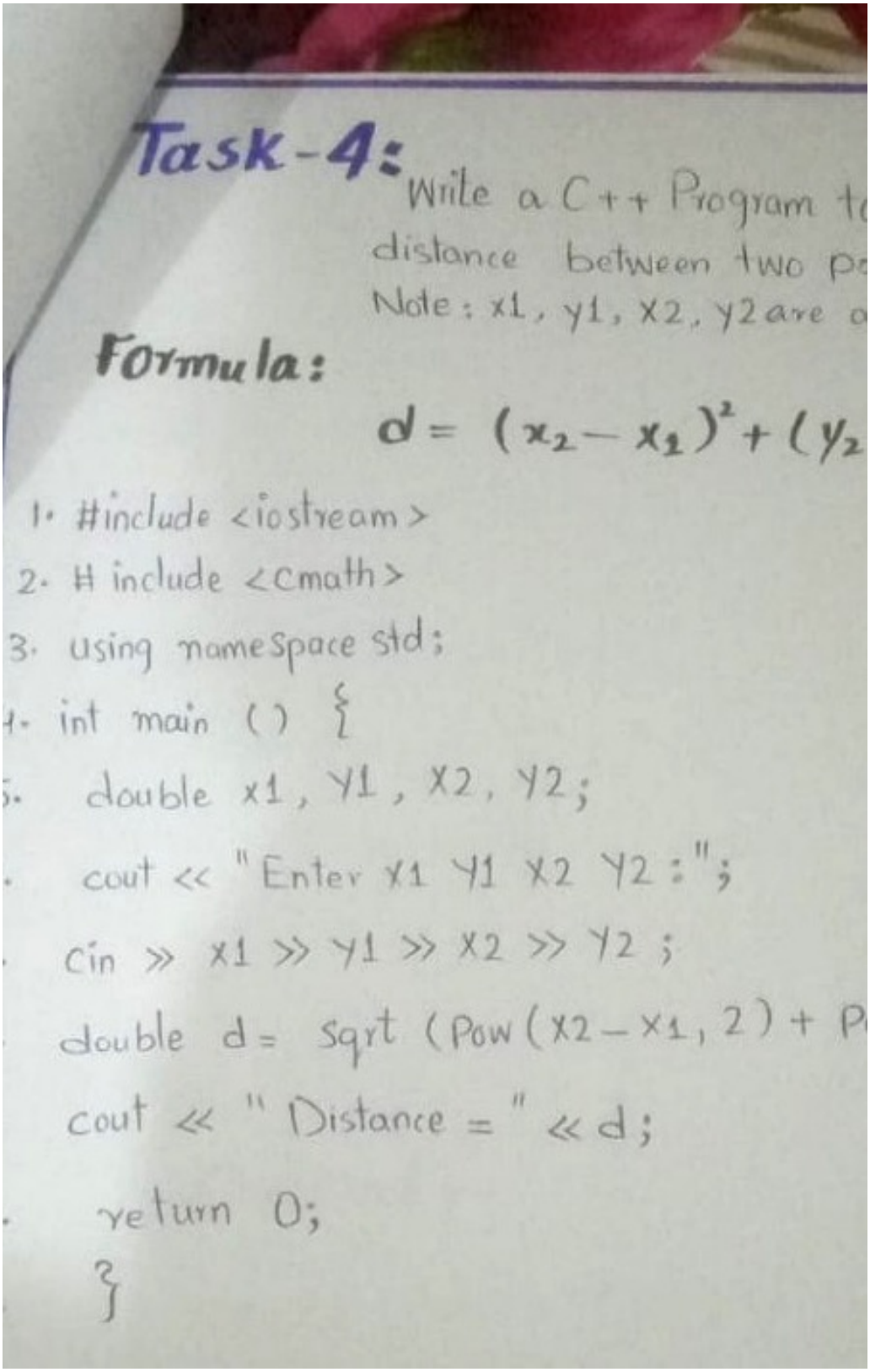
    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
```

Output

```
Please enter your 3 numbers: 10 20 30
10
20
30
30
20
10
```

Task 4 — Distance Between Two Points

Write a C++ program to calculate the distance between two points using the given formula.



Task-4: Write a C++ Program t
distance between two p
Note: x1, y1, x2, y2 are

Formula:

$$d = (x_2 - x_1)^2 + (y_2$$

```
1. #include <iostream>
2. # include <cmath>
3. using namespace std;
4. int main () {
5.   double x1, Y1, X2, Y2;
.    cout << "Enter X1 Y1 X2 Y2 :";
.   Cin >> x1 >> y1 >> X2 >> Y2 ;
.   double d = sqrt (Pow (X2 - X1, 2) + P
    cout << " Distance = " << d;
.    return 0;
    }
```

Task 4 — Code & Output

Code in Compiler

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2, distance;
    cout << "Enter x1 y1 x2 y2: ";
    cin >> x1 >> y1 >> x2 >> y2;

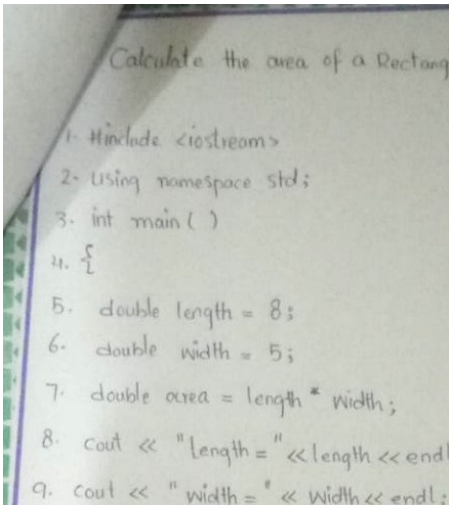
    distance = sqrt(pow(x2-x1, 2) + pow(y2-y1, 2));
    cout << "Distance = " << distance << endl;
    return 0;
}
```

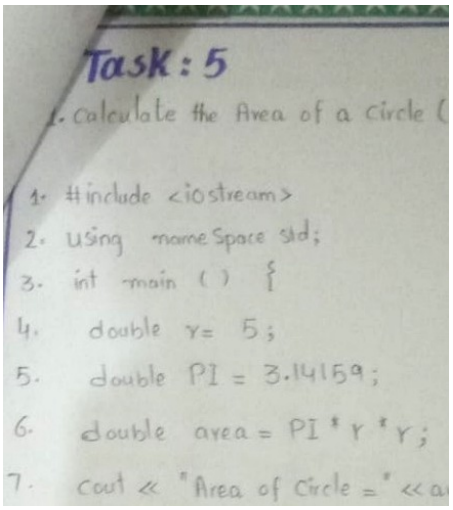
Output

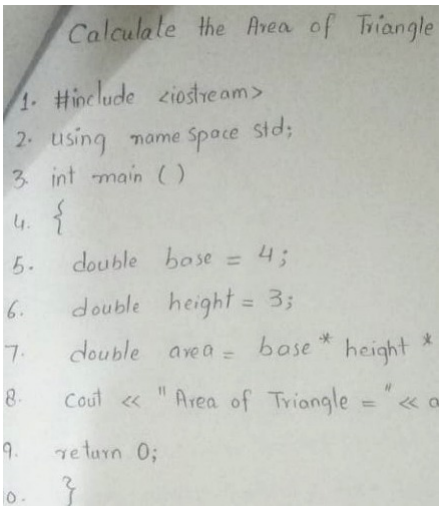


Task 5 — Areas of Shapes

Calculate the area of a rectangle, circle and triangle. The handwritten code crops are arranged in this order.







Task 5 — Code & Output

Code in Compiler

```cpp
#include <iostream>
using namespace std;

int main()
{
    double r, length, width, base, height;
    const double PI = 3.14159;

    cout << "Enter radius of circle: ";
    cin >> r;
    cout << "Circle area = " << PI * r * r << endl;

    cout << "Enter length and width: ";
    cin >> length >> width;
    cout << "Rectangle area = " << length * width << endl;

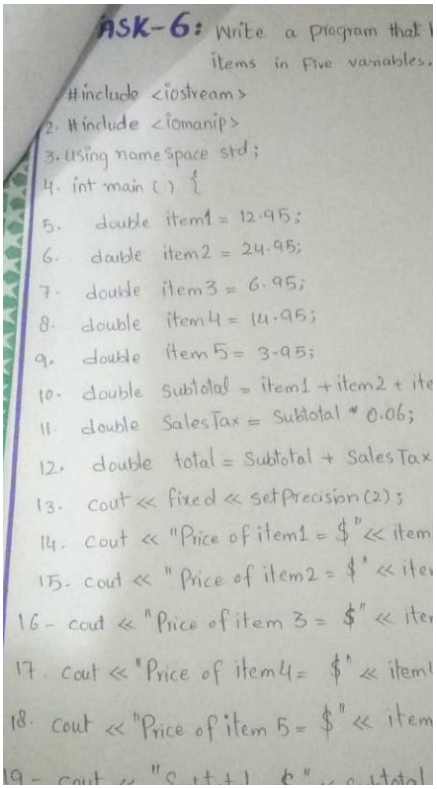
    cout << "Enter base and height: ";
```

Output

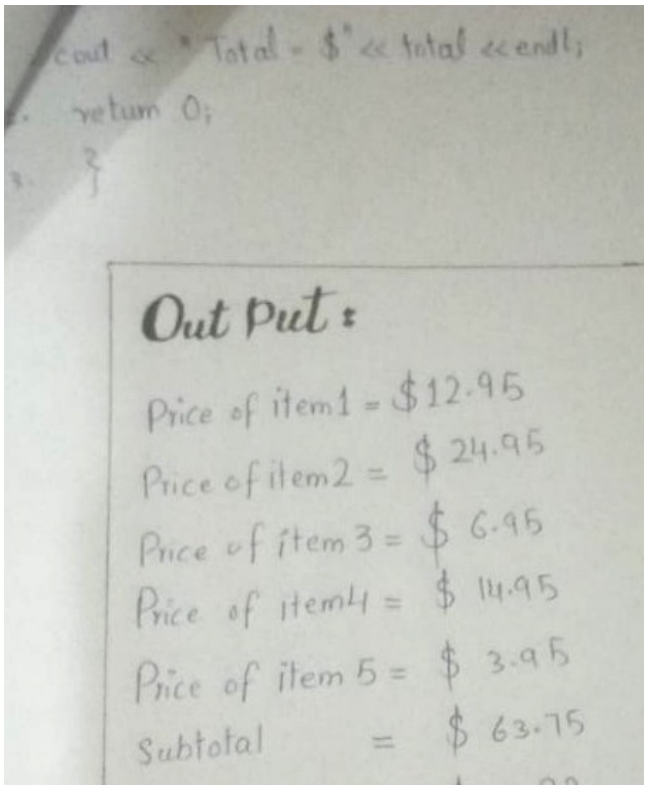
```
Enter radius of circle: 3
Circle area = 28.2743
Enter length and width: 5 4
Rectangle area = 20
Enter base and height: 6 3
Triangle area = 9
```

Task 6 — Prices of Five Items

Store the prices of five items, calculate the subtotal, 6% sales tax and total.



```
ASK-6: Write a program that
       items in five variables.
#include <iostream>
#include <iomanip>
using namespace std;
int main () {
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;
    double Subtotal = item1 + item2 + ite
    double SalesTax = Subtotal * 0.06;
    double total = Subtotal + SalesTax
    cout << fixed << setPrecision(2);
    cout << "Price of item1 = $" << item
    cout << "Price of item2 = $" << ite
    cout << "Price of item 3 = $" << ite
    cout << "Price of item4 = $" << item
    cout << "Price of item 5 = $" << item
```



```
    cout << "Total = $" << total << endl;
    return 0;
}
```

Out Put :

```
Price of item1 = $12.95
Price of item2 = $ 24.95
Price of item 3 = $ 6.95
Price of item4 = $ 14.95
Price of item 5 = $ 3.95
Subtotal = $ 63.75
```

Task 6 — Code & Output

Code in Compiler

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    double item1 = 12.95, item2 = 24.95, item3 = 6.95;
    double item4 = 14.95, item5 = 3.95;
    double subtotal = item1 + item2 + item3 + item4 + item5;
    double tax = subtotal * 0.06;
    double total = subtotal + tax;

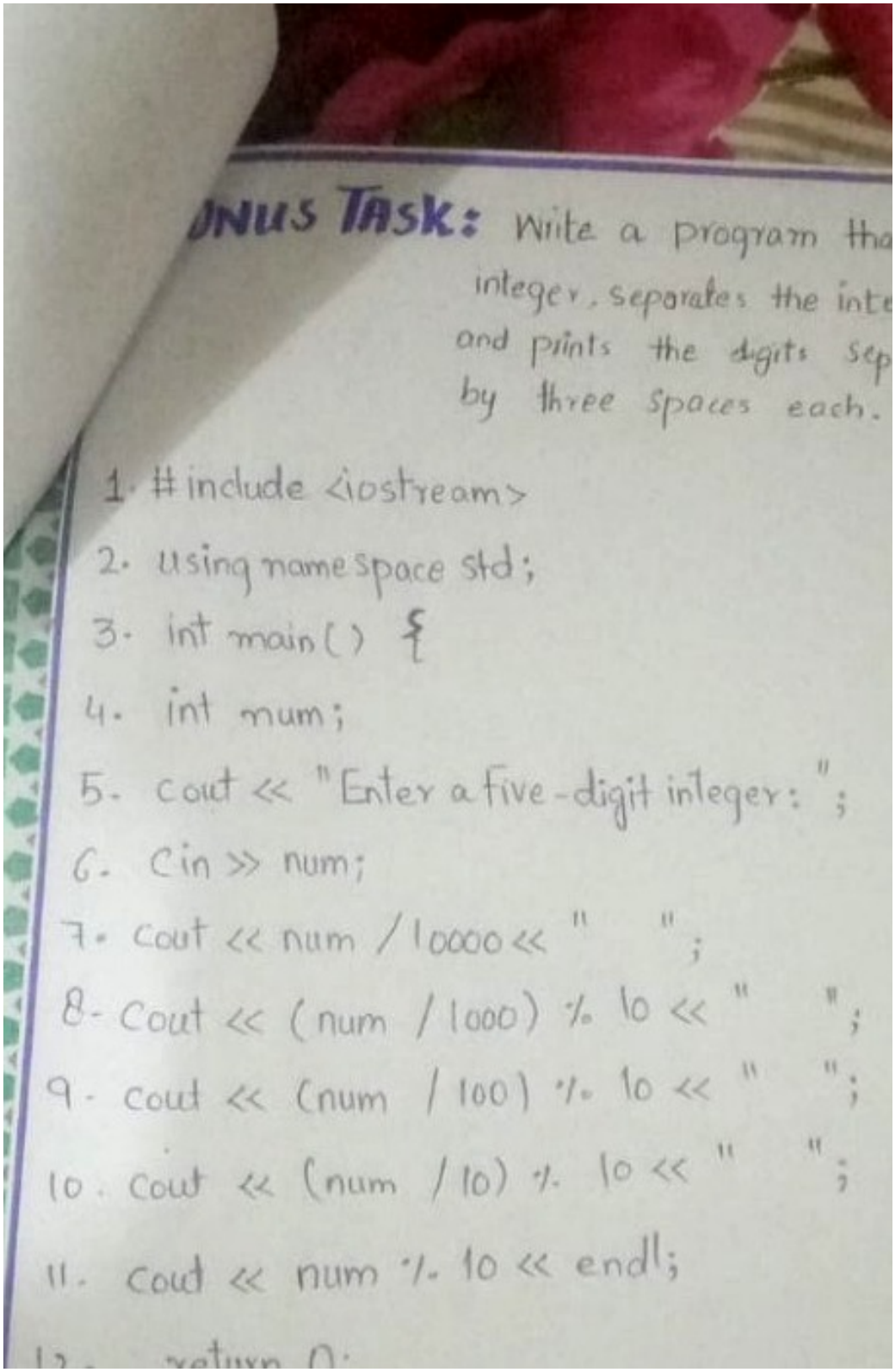
    cout << fixed << setprecision(2);
    cout << "Item 1: $" << item1 << endl;
    cout << "Item 2: $" << item2 << endl;
    cout << "Item 3: $" << item3 << endl;
    cout << "Item 4: $" << item4 << endl;
```

Output

```
Item 1: $12.95
Item 2: $24.95
Item 3: $6.95
Item 4: $14.95
Item 5: $3.95
Subtotal: $63.75
Sales tax (6%): $3.83
Total: $67.58
```

Bonus Task — Separate Five Digits

Input a five-digit integer and print its digits separated by spaces.



Bonus Task — Code & Output

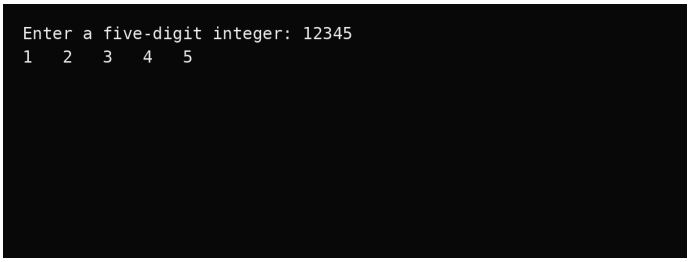
Code in Compiler

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cout << "Enter a five-digit integer: ";
    cin >> n;

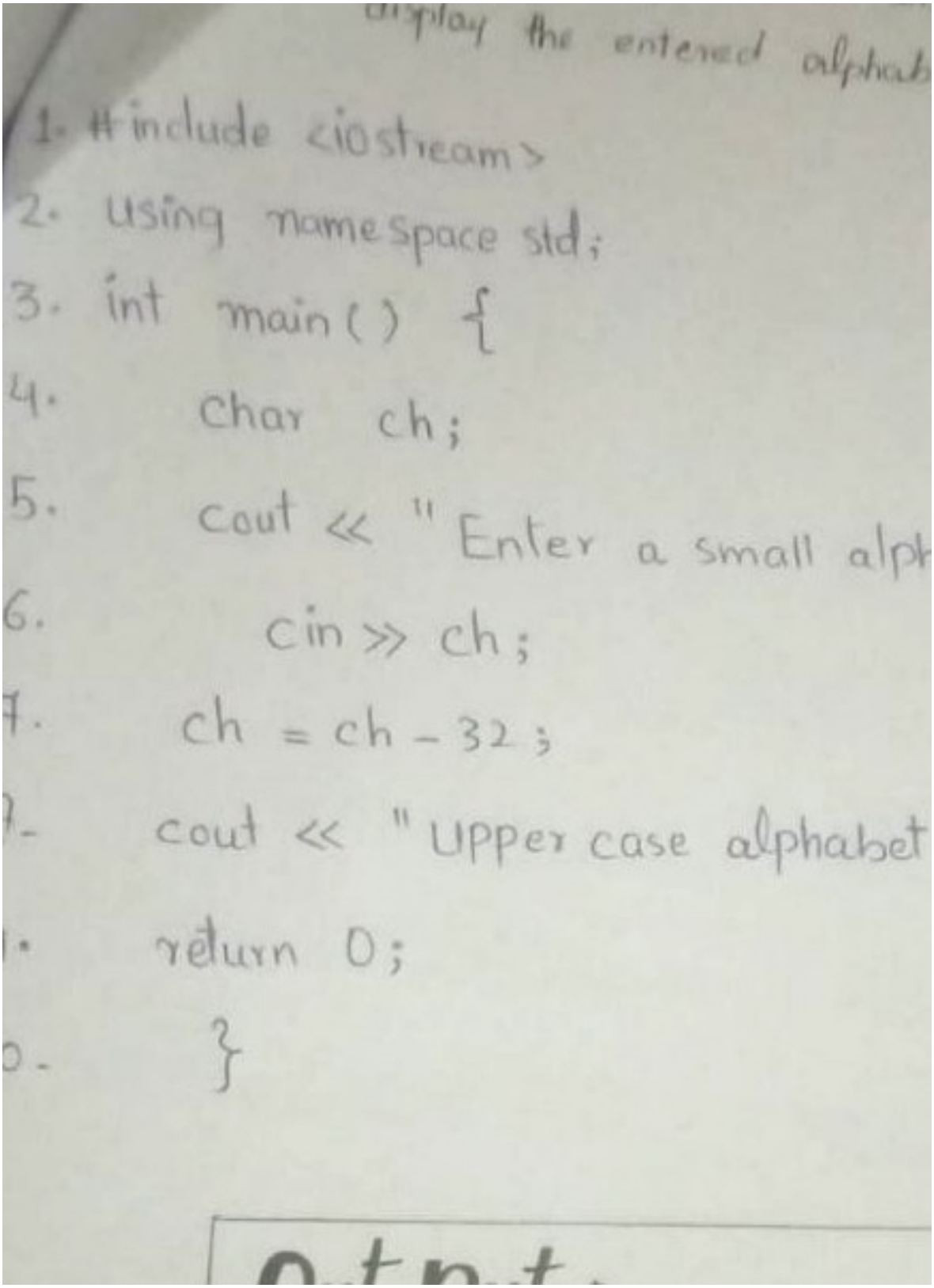
    cout << n / 10000 << "   ";
    cout << (n / 1000) % 10 << "   ";
    cout << (n / 100) % 10 << "   ";
    cout << (n / 10) % 10 << "   ";
    cout << n % 10 << endl;
    return 0;
}
```

Output



Additional Task — Uppercase Alphabet

Prompt the user for a lowercase alphabet and display the corresponding uppercase alphabet.



```
display the entered alphab
1. #include <iostream>
2. using namespace std;
3. int main() {
4.     char ch;
5.     cout << "Enter a small alph
6.         cin >> ch;
7.     ch = ch - 32;
8.     cout << "Upper case alphabet
9.     return 0;
10.    }
```

Additional Task — Code & Output

Code in Compiler

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;
    cout << "Enter a lowercase alphabet: ";
    cin >> letter;
    letter = letter - 32;
    cout << "Uppercase alphabet: " << letter << endl;
    return 0;
```

Output

